BINOMIAL THEOREM

Expanding a power of a difference

A difference is a sum with a negative second summand: the coefficients are still a row of Pascal's triangle, and the signs come from the powers of $-3b$.

01

POWER OF A DIFFERENCE — BEFORE

$$(2a - 3b)^3$$

↓ Substitute into row 3 — bases 2a and −3b, coefficients 1, 3, 3, 1

02

TERM BY TERM

$$(2a)^3 + 3(2a)^2(-3b) + 3(2a)(-3b)^2 + (-3b)^3$$

↓ Simplify each term — work out each power of −3b, sign included, before multiplying

03

EXPANDED POLYNOMIAL

AFTER

$$8a^3 - 36a^2b + 54ab^2 - 27b^3$$

▼ WHY +54AB²?